THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **Introduction to Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	highest power of the variable is 2
2	**(a)** 4	highest power of the variable is 4
3	**(a)** -5	$-4(3)+(7)=-5$
4	**(d)** 81	substitute $x=-3$ to get 81
5	**(b)** $2n+3$	common difference 2, so $a_n=2n+3$
6	**(c)** 2 and 5	compare with $y=ax+b$
7	**(d)** -1 and -2	compare with $y=ax+b$
8	**(a)** lies on the line	RHS at $x=3$ is 1, equal to 1
9	**(c)** −10	term without the variable
10	**(d)** falls (negative slope)	slope $-2<0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_n=3n+(1)$; $a_{10}=3(10)+(1)=\mathbf{31}$	1+1
12	$a_n=5n+(-3)$; $a_{12}=5(12)+(-3)=\mathbf{57}$	1+1
13	$a_n=-5n+(105)$; $a_8=-5(8)+(105)=\mathbf{65}$	1+1
14	$a_n=4n+(3)$; $a_{15}=4(15)+(3)=\mathbf{63}$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let Salil's age $=x$, mother $=3x$. $(x+5)+(3x+5)=70\Rightarrow 4x=60\Rightarrow x=15$. Salil 15 years, mother 45 years.	1+1+1
16	Let the numbers be $2k$ and $5k$: $5k-2k=63\Rightarrow k=21$. Numbers: 42 and 105.	1+1+1
17	₹5 coins $=x$, ₹2 coins $=3x$. $5x+6x=88\Rightarrow x=8$. So 8 five-rupee coins and 24 two-rupee coins.	1+1+1
	OR: $x+4x=300\Rightarrow x=60$. Pieces: 60 ft and 240 ft.	1+1+1
18	Width $=w$, length $=2w+3$. $2(3w+3)=24\Rightarrow w=3$. Width 3 cm, length 9 cm.	1+1+1
	OR: $120-9n$; after 7 hours: $120-63=57$.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $C(5)=400$, $C(15)=1000$. (ii) $100+60d=700\Rightarrow d=10$ km. (iii) Linear growth; slope 60 = cost per km, intercept 100 = fixed charge.	2+1+2
	OR: (i) e.g. $y=2x-1$: $(0,-1),(1,1)$; $y=2x+1$: $(0,1),(1,3)$; $y=2x+5$: $(0,5),(1,7)$. (ii) All parallel (same slope 2); they shift up/down as b changes. (iii) $(0,5)$.	3+1+1
20	(i) $2n-1$. (ii) 29 and 51. (iii) $2n-1=21\Rightarrow n=11$; $n=24$ for 47. (iv) Slope 2 = constant difference between consecutive terms.	1+1+2+1
	OR: (i) $350=10a+b$, $550=20a+b$ ⇒ $a=20$, $b=150$. (ii) $y=20x+150$; $x=35$: ₹850. (iii) ₹20 per GB and ₹150 fixed monthly fee.	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Introduction to Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	highest power of the variable is 3
2	**(c)** 7	$5(2)+(-3)=7$
3	**(b)** -17	$6(-2)+(-5)=-17$
4	**(d)** 5	substitute $x=2$ to get 5
5	**(c)** $-3n+13$	common difference -3, so $a_n=-3n+13$
6	**(a)** -3 and 1	compare with $y=ax+b$
7	**(b)** 5 and 0	compare with $y=ax+b$
8	**(d)** does not lie on the line	RHS at $x=1$ is 7, not equal to 8
9	**(d)** the origin	$x=0 \Rightarrow y=0$
10	**(a)** 7	$b=4$, $a=3 \Rightarrow a+b=7$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	At $x=0$: $5(0)+(-3)=-3$; at $x=-1$: $5(-1)+(-3)=-8$	1+1
12	At $x=2$: $3(2)+(4)=10$; at $x=-2$: $3(-2)+(4)=-2$	1+1
13	At $x=1$: $-2(1)+(6)=4$; at $x=4$: $-2(4)+(6)=-2$	1+1
14	At $x=3$: $4(3)+(-1)=11$; at $x=0$: $4(0)+(-1)=-1$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$400=10a+b$, $500=14a+b$. Subtract: $4a=100 \Rightarrow a=25$; $b=400-250=150$. $y=25x+150$.	1+1+1
16	$800=10a+b$, $1100=15a+b \Rightarrow a=60,\ b=200$. $y=60x+200$; for $x=20$: ₹1400.	2+1
17	$0=32a+b$, $100=212a+b \Rightarrow 180a=100$, $a=\frac{5}{9}$, $b=-\frac{160}{9}$. °C $=\frac{5}{9}(^\circ F-32)$. At 98.6 °F: $\frac{5}{9}\times 66.6=37$ °C.	1+1+1
	OR: $C(15)=100+900=$ ₹1000. $100+60d=700 \Rightarrow d=10$ km. Positive slope 60 ⇒ linear growth.	1+1+1
18	$h=1.75+0.5t$; $h(7)=5.25$ ft. The height rises by the same 0.5 ft each month (constant positive slope).	1+1+1
	OR: $v=10000-800t$; $v(3)=7600$. $10000-800t=4000 \Rightarrow t=7.5$ years.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x+(x+10)=64 \Rightarrow x=27$; numbers 27, 37. (ii) $3x+5+x+5=70 \Rightarrow x=15$; son 15, mother 45.	2+3
	OR: (i) t: 0,1,2,3,4,5,6 → h: 3, 2.5, 2, 1.5, 1, 0.5, 0. (ii) Linear decay — constant decrease of 0.5 m per month. (iii) At $t=6$ months. (iv) Straight line through $(0,3)$ and $(6,0)$; slope -0.5, h-intercept 3.	1+1+1+2
20	$a=\frac{5}{9}$, $b=-\frac{160}{9}$. (i) $\frac{5}{9}(77-32)=25$ °C. (ii) $25=\frac{5}{9}(F-32) \Rightarrow F=77$ °F.	3+1+1
	OR: (i) ₹7,600. (ii) 10000, 9200, 8400, 7600, 6800, 6000, 5200, 4400, 3600. (iii) $v=10000-800t$; constant decrease ₹800 per year. (iv) $10000-800t=2000 \Rightarrow t=10$ years.	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.